CUONG T. PHAM

Ph.D. Student | Graduate Research Assistant
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Research Interests

Structural Engineering; Structural Damage Assessment; Coastal Flood Risk and Natural Hazards; Machine Learning and Deep Learning; Computer Vision; Structural Vulnerability; Structural Dynamics.

Education

University of South Alabama - Mobile, Alabama, USA 2024–present
Ph.D. in Systems Engineering; expected May 2028.

University of South Alabama - Mobile, Alabama, USA May 2026
Master of Science in Civil Engineering; GPA 4.0/4.0.

University of Da Nang – University of Science and Technology (UD–DUT) - Vietnam May 2022
Bachelor of Engineering in Civil Engineering; GPA 3.61/4.0; ranked 1/120 (Valedictorian).

Peer-Reviewed Publications

1. Umeike, R., Dao, T., Crawford, S., van de Lindt, J., Johnston, B., (Lisa) Wang, W., Do, T., Mofikoya, A., Banjara, S. & **Pham, C.** ASTRA: Architectural inSights for post-ToRnado damAge recognition. *Expert Systems with Applications* **333.** DOI: 10.1016/j.eswa.2026.133915 (2027).
2. Do, V. H., **Pham, T. C.,** Phan, H. N., Nguyen, M. H. & Huynh, P. N. Deep learning for bridge component classification and damage detection from UAV imagery. *Proceedings of the Institution of Civil Engineers - Bridge Engineering* **179,** 194–205. DOI: 10.1680/jbren.25.00006 (2025).
3. Huynh, P. N., Nguyen, M. H., Do, V. H., Phan, H. N., **Pham, T. C.** & Nguyen, V. H. Investigation of the application scope of bond strength empirical formulas for various FRP bars in concrete: A case study utilizing the safety probability value with a large test dataset. *Case Studies in Construction Materials* **20.** DOI: 10.1016/j.cscm.2024.e03168 (2024).

Manuscripts Under Review and Submitted

1. **Pham, C. T.**, Do, T. Q., and Webb, B. M. *First-Floor Height Estimation for Elevated Buildings Using a Deep Learning-Based Object Detection Method*. *Journal of Computing in Civil Engineering* (ASCE). Revised manuscript submitted (2026).
2. **Pham, C. T.**, Do, T. Q., and Webb, B. M. *Coastal Flood Damage and Loss Assessment under Storm Surge and Sea-Level Rise with AI-Derived First-Floor Elevation*. *Journal of Structural Engineering* (ASCE). Submitted October 2026.
3. **Pham, C. T.** and Do, T. Q. *Machine Learning and Deep Learning in Coastal Flood Management: A Systematic Review on Data, Methods, Applications, and Future Directions*. *Natural Hazards* (Springer Nature). Submitted October 2026.

Preprints

1. Umeike, R., **Pham, C.,** Hausen, R., Dao, T., Crawford, S., Brown-Giammanco, T., Lemson, G., van de Lindt, J., Johnston, B., Mitschang, A. & Do, T. *TornadoNet: Real-Time Building Damage Detection with Ordinal Supervision* 2026. arXiv: 2603.11557 [cs.CV]. https://arxiv.org/abs/2603.11557.

Peer Review Experience

Magazine of Concrete Research April 2026

Peer reviewer; 1 completed review.

Proceedings of the Institution of Civil Engineers – Construction Materials August 2026

Peer reviewer; 1 completed review.

Research Experience

Graduate Research Assistant May 2024–Present

University of South Alabama

- Developed computer vision and geometric methods to estimate first-floor heights of elevated residential buildings using street-view imagery.
- Conducted coastal flood damage and economic loss assessments under storm surge and sea-level rise scenarios using the IN-CORE platform.
- Conducted a systematic review of machine learning and deep learning applications in coastal flood management.
- Collaborated on deep learning-based post-tornado building damage assessment using street-view imagery, including object detection, damage severity classification, and model performance evaluation.
- Conducting experimental wave flume research to investigate the dynamic response and structural vulnerability of elevated wood-frame buildings subjected to coastal wave loading.

Research and Teaching Assistant 2022–2024

University of Da Nang – University of Science and Technology

- Applied artificial neural networks to structural reliability analysis of steel beams.
- Evaluated empirical bond-strength models for FRP reinforcement in concrete using statistical and reliability-based approaches.
- Assisted with undergraduate teaching in structural engineering and related courses.

Undergraduate Researcher 2021–2022

University of Da Nang – University of Science and Technology

- Collected UAV imagery and developed an annotated dataset for bridge component classification and structural damage detection.
- Developed and trained convolutional neural network models using the collected dataset for automated bridge damage detection.

Selected Conference Presentations

1. **Pham, C. T.**, Do, T. Q., and Webb, B. M. *Integrating AI in Flood Risk and Loss Estimation under Storm Surge and Sea Level Rise Scenarios: A Case Study for Dauphin Island, Alabama*. Oral presentation, 2026 World Environmental & Water Resources Congress (EWRI), Mobile, Alabama, USA, April 29, 2026.

2. **Pham, C. T.** *Integrating AI in Flood Risk and Loss Estimation under Storm Surge and Sea Level Rise Scenarios: A Case Study for Dauphin Island, Alabama*. Oral presentation, 2nd Annual Shelby Hall Graduate Research Forum, University of South Alabama, Mobile, Alabama, USA, March 27, 2026.

3. **Pham, C. T.**, Do, T. Q., and Webb, B. M. *Dauphin Island Testbed for Multi-Hazard Risk Assessment and Loss Analysis for Projected Sea Level Rise Scenarios*. Poster presentation, 7th Global Summit of the Global Alliance of Disaster Research Institutes (GADRI), Colorado State University, Fort Collins, Colorado, USA, July 21–23, 2025.

4. **Pham, C. T.**, Do, V. H., Nguyen, M. H., and Phan, H. N. *Unmanned Aerial Vehicles and Machine Learning Techniques to Detect Bridge Damage*. Oral presentation, Joint International Conference on Environment, Earth Science and Sustainability (ICES), Ho Chi Minh City, Vietnam, 2022.

Honors, Scholarships, and Awards

SEI Student Scholarship to Structures Congress 2025
Structural Engineering Institute (SEI), ASCE.

NHERI Strategic Program to Accelerate Researchers in Computing (SPARC) 2025
Natural Hazards Engineering Research Infrastructure (NHERI).

NHERI Computational Academy 2024
Natural Hazards Engineering Research Infrastructure (NHERI).

Valedictorian – Faculty of Road and Bridge Engineering 2022
University of Da Nang – University of Science and Technology (ranked 1st of 120 graduates).

First Prize – Student Scientific Research Conference 2022
Faculty of Road and Bridge Engineering, UD-DUT.

Second Prize – Student Scientific Research Conference 2022
University of Da Nang – University of Science and Technology.

Promising Undergraduate Research Project Award 2022
University of Da Nang – University of Science and Technology.

Odon Vallet Scholarship for Academic Excellence 2019
Odon Vallet Scholarship Program, Vietnam.

Technical Skills

Programming and Data Analysis: Python, MATLAB, C++; statistical and numerical analysis; machine learning and deep learning.

Computational Fluid Dynamics (CFD): OpenFOAM, ANSYS Fluent; numerical simulation of fluid flow and wave dynamics.

Structural Engineering: SAP2000, MIDAS Civil, AutoCAD, Revit; structural analysis and damage assessment.

Computer Vision and Experimental Research: UAV image analysis; object detection; experimental data acquisition and analysis; structural dynamics.

Professional Qualifications

Fundamentals of Engineering (FE) Exam – Passed August 2026 (NCEES).